INCH-POUND

MIL-PRF-49471/3A(CR)
30 November 2000
Superseding
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2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X590/U

This specification is approved for use within the Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

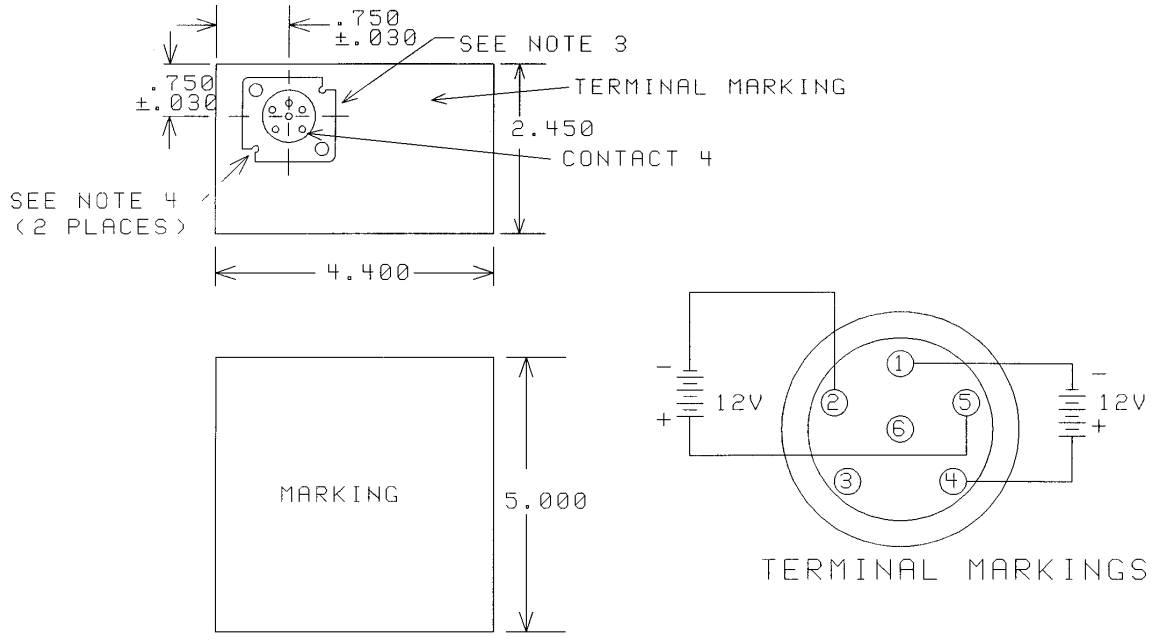

Figure 1. Battery dimensions and terminal markings.

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -0.063, +0.000.

3. The battery terminal shall consist of a miniature circular battery connector, Dwg SC-C-179492, except as noted herein.

4. Connector socket body configuration shall accept two (2) each 0.156 inch minimum diameter guideposts as shown. Depth of cavity shall be 0.657 inch minimum.

5. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 0.0625 inch below adjacent outside of the jacket when used initially, during, and after subsequent insertions of the mating plug during testing.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 33 volts in 24 volt mode
16.5 volts in 12 volt mode

Terminals: Six (6) hole socket type with no obstruction of any holes (see notes 3 and 5).

Weight (maximum): 3.5 pounds (1588 grams).

Cut-off voltage for capacity: 20.00 volts in 24 volt mode
10.00 volts in 12 volt mode

Charge protection: Applicable.

State of Charge Indicator: Applicable

Battery short circuit test: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

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Capacity Test*	Minimum Capacity Requirements in Hours to 20.00 volts in Series, 10.00 volts in Parallel
I & IP	3.4
L & LP	2.2
H & HP	5.6
HT & HTP	5.4
LT & LTP	2.0
IT & ITP	3.0
LR1	48.0

*NOTE: Sample batteries that are to be assigned to a capacity test (except for the LR1) shall be equally divided between the designated capacity test and the capacity test ending in P (e.g. I and IP). The odd battery, if present, is assigned to parallel discharge capacity test. For the capacity tests that do not have the P's, the battery's legs shall be discharged in series. For the capacity tests that have the P's, the battery's legs shall be discharged in parallel.

Initial voltage delay: During the I, H, HT, and IT discharge tests of the two (2) 12 volt sections connected in series under the designated loads, initial voltage below the 20.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 20.00 volts cannot exceed a 60 second duration. During the IP, HP, HTP, and ITP discharge tests of the two (2) 12 volt sections connected in parallel under the designated loads, initial voltage below the 10.00 volt minimum voltage cannot exceed a 5.0 second duration. During LP and LTP discharge tests initial voltage below 10.00 volts cannot exceed a 60 second duration.

Connector: The connector configuration and operating temperature range requirements shall be certified in the First Article Test Report. The connector shall be in accordance with SC-C-179492 with the following exceptions: (1) Sheet two of SC-C-179492 does not apply. (2) Mounting screws are optional. (3) Operating temperature range for the connector shall be from a low of-55°C to a high of 125°C. (4) The connector test described in paragraph 4.7.15 shall be performed 10 times with SM-D-687888. (5) Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin.

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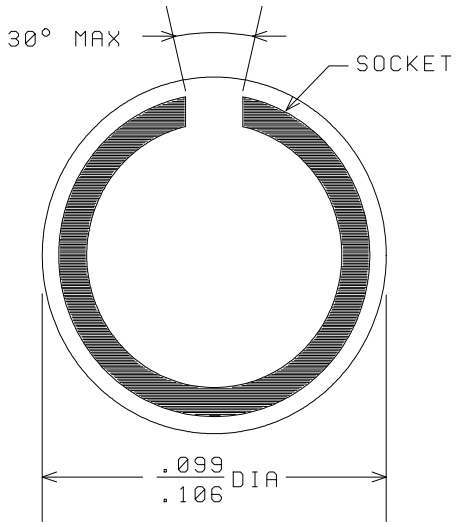

Figure 2. Socket (Top View).

METHOD OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all specified capacity tests are described in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2.0 volts. Each discharged cell shall then be connected in series to nine fresh cells. Each string shall be discharged at 2.0 amperes at 70±5°F for 3.4 hours.

Batteries: For all capacity tests that do not end in P, except the LR1, H and HT, the batteries shall be discharged in the 24 volt mode by connecting pins 2 & 4 and discharged between pins 1 & 5 at 2.0 amperes constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests the batteries shall be discharged in the 24 volt mode as described above, at 2.0 amperes for five minutes, followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged in the 24 volt mode as described above at 145 milliamperes to 20.5 volts, then at 5.5 watts to 3 volts.

Forced discharge: After discharge to zero volts on the I capacity test, the batteries shall be force discharged in the 24 volt mode as described above at 2.0 amperes for a

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minimum of 10 minutes. After discharge to zero volts on the IP capacity test, batteries shall be force discharged with the two 12 volt sections connected in parallel at 4.0 amperes constant current for a minimum of 10 minutes.

Overload test: Batteries shall be discharged in the 24 volt mode as described above at 4 amperes.

Abuse test predischarge: Designated samples shall be discharged in the 24 volt mode as described above at 2.0 amperes for 1.7 hours.

Abuse test, pulse discharge: Batteries shall be discharged in the 24 volt mode at 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to cut-off voltage.

For all capacity tests that end in P, except the HP and HTP, the batteries shall be discharged with the two 12 volt sections connected in parallel at 4.0 amperes constant current to cut-off volts (zero volts for the IP test). For the HP and HTP capacity tests the batteries shall be discharged with the two 12 volt sections in parallel at 4.0 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts.

Closed-circuit voltage (Battery and cell string tests): Load resistance of 5.0 ohms shall be used between socket holes 1 & 4 and 2 & 5. Voltage shall be 10.0 volts or greater within ten seconds.

Cell closed circuit test: Load resistance of 1 ohm shall be used. Voltage shall be above 2.0 volts in 5 seconds.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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